$\therefore f(A,B,C,D) = CD + A\bar{B}CD + \bar{A}B\bar{C}D$ (Ans)

C.W

26.10.2026

Minterm & Maxterm:

• Minterm:

A	B	C	Minterm	
0	0	0	$\bar{A}\bar{B}\bar{C}$	m_0
0	0	1	$\bar{A}\bar{B}C$	m_1
0	1	0	$\bar{A}B\bar{C}$	m_2
0	1	1	$\bar{A}BC$	m_3
1	0	0	$A\bar{B}\bar{C}$	m_4
1	0	1	$A\bar{B}C$	m_5
1	1	0	$AB\bar{C}$	m_6
1	1	1	ABC	m_7

Maxterm:

A	B	C	Maxterm	
0	0	0	$\bar{A}+\bar{B}+\bar{C}$	M_0
0	0	1	$\bar{A}+\bar{B}+C$	M_1
0	1	0	$\bar{A}+B+\bar{C}$	M_2
0	1	1	$\bar{A}+B+C$	M_3
1	0	0	$A+\bar{B}+\bar{C}$	M_4
1	0	1	$A+\bar{B}+C$	M_5
1	1	0	$A+B+\bar{C}$	M_6
1	1	1	$A+B+C$	M_7

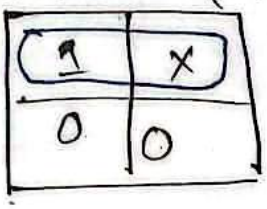
(•) K-map:

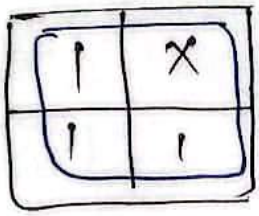
Don't care condition:

- Function that have unspacified output for some input combinations are called incompletely specified functions. Unspecified minterems of function are called don't whether the value 0 or 1 assigned to "F" for a particular minterem. Don't care conditions are

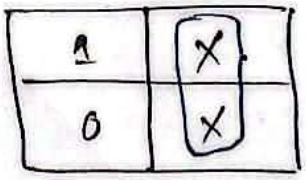
represented by 'X' in the K-map table.

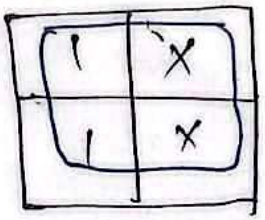
• possible: (চলবে)





• Not possible: (চলবে না)





HW:

1. Using K-map simplify the following function:

$F(A,B,C,D) = \sum m(1,3,4,6,8,9,11,13,15) + d(0,2,$